Systems of Linear Equations

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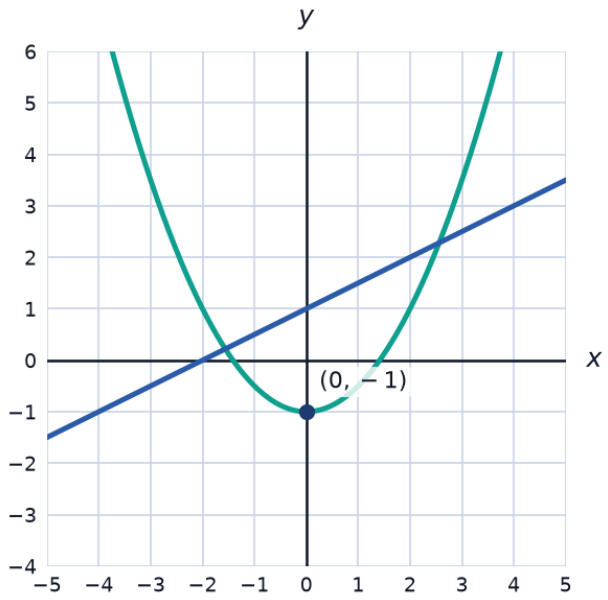
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Setting up the augmented matrix

Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).

2 [Foundational] • Target: Setting up the augmented matrix

Write the augmented matrix for the system $x - y = 4$, $3x + 2y = 7$.



3 [Foundational] • Target: Elementary row operations

List the three types of elementary row operations.

4 [Foundational] • Target: Elementary row operations

Given the augmented matrix with rows $\langle 1, 2, 5\rangle$, $\langle 3, -1, 1\rangle$, apply the row operation $R_2 \to R_2 - 3R_1$ and give the new Row 2.

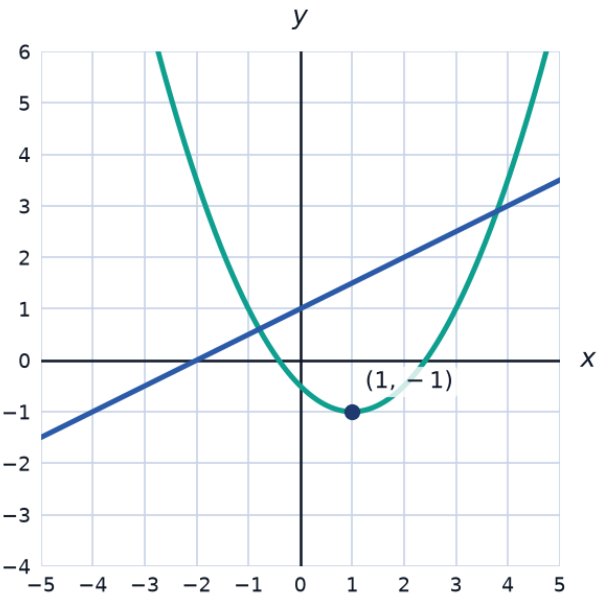
5 [Foundational] • Target: Solving 2x2 systems by elimination

Solve $2x + 3y = 7$, $x - y = 1$.

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6 [Foundational] • Target: Solving 2x2 systems by elimination

Solve $3x - 2y = 4$, $5x + y = 11$.



7 [Foundational] • Target: Solving 2x2 systems by elimination

Solve $4x + y = 10$, $2x - 3y = -8$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.

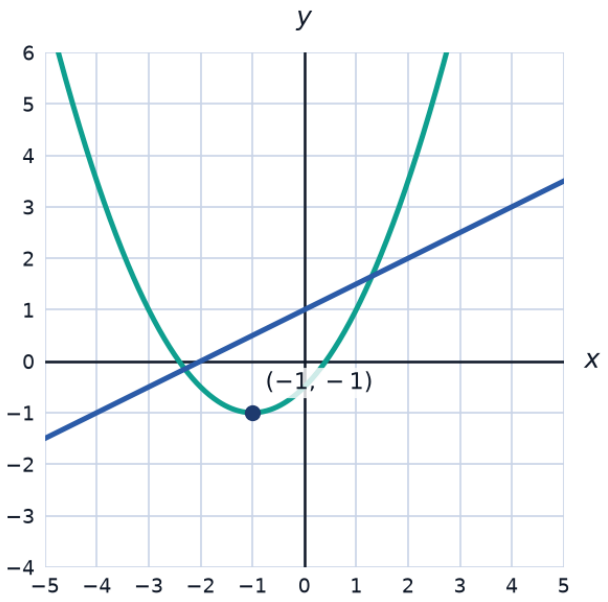
9 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Solve $x + 2y + z = 4$, $y - z = 1$, $3x - y + 2z = 1$.

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10 [Intermediate] • Target: Row echelon form and back-substitution

A system's augmented matrix has been reduced to row echelon form with rows $\langle 1, 2, -1, 4\rangle$, $\langle 0, 1, 3, -2\rangle$, $\langle 0, 0, 1, 5\rangle$ (variables x, y, z). Use back-substitution to solve.



11 [Intermediate] • Target: Row echelon form and back-substitution

Explain why row echelon form makes a system easy to solve, in one or two sentences.

12 [Intermediate] • Target: Inconsistent and dependent systems

Solve (or classify) the system $x + y = 3$, $2x + 2y = 8$.

13 [Intermediate] • Target: Inconsistent and dependent systems

Solve (or classify) the system $x + y = 3$, $2x + 2y = 6$.

14 [Intermediate] • Target: Inconsistent and dependent systems

For the system $x - 2y = 4$, $-3x + 6y = k$, find the value of k that makes the system have infinitely many solutions (rather than no solution).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

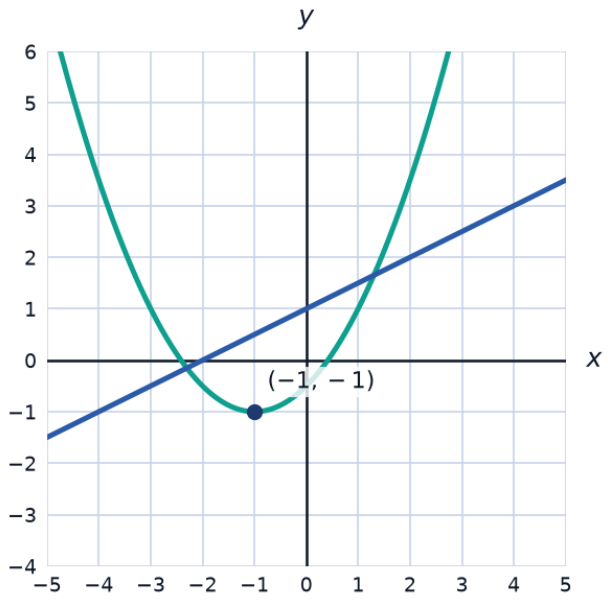
15 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 1.01 Vectors in n Dimensions

Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.

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16 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Why is the restriction $c \neq 0$ needed for this row operation?



17 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 4.01 Determinants

Explain briefly why adding a multiple of one row to another row also preserves the solution set.

18 [Challenging] • Target: Systems with a parameter • Cross-Lesson: 1.01 Vectors in n Dimensions

For what value(s) of k does the system $x + ky = 3$, $2x + 4y = 6$ have infinitely many solutions?

19 [Challenging] • Target: Systems with a parameter • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For what value of k does the system $2x - y = 5$, $4x + ky = 1$ have NO unique solution (i.e., the coefficient matrix is singular)?

20 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 4.01 Determinants

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 L of a 15% solution. Set up the system of equations (let x, y be the liters of each solution used).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Augmented Matrices and Row Echelon Form (REF)

Which matrix represents a valid Reduced Row Echelon Form (RREF)?

a. $[[1, 0, 3], [0, 1, -2], [0, 0, 0]]$

b. $[[1, 2, 3], [0, 1, -2], [0, 0, 0]]$

c. $[[2, 0, 0], [0, 1, 0], [0, 0, 1]]$

d. $[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$

22 [Foundational] • Target: Augmented Matrices and Row Echelon Form (REF)

How many pivots does a 3×5 matrix have at most?

a. 3

b. 5

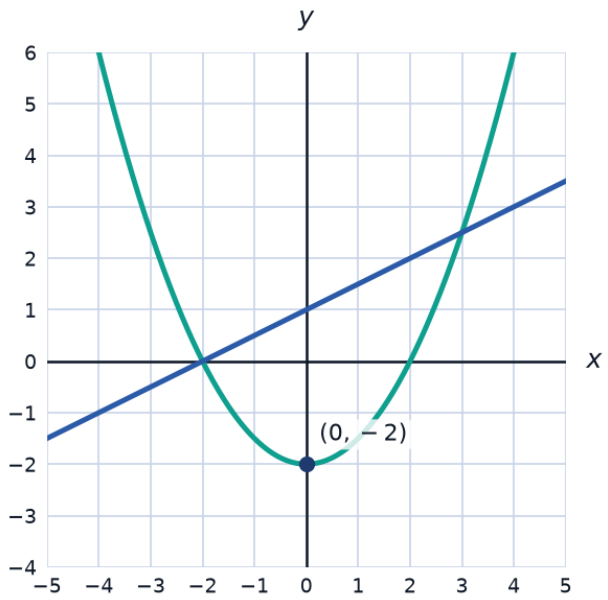
c. 2

d. 15

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23 [Foundational] • Target: Augmented Matrices and Row Echelon Form (REF)

An augmented matrix has a row of the form $[0, 0, \ldots, 0 \mid b]$ with $b \neq 0$. What does this establish about the linear system?



a. The system is inconsistent (has no solution)

b. The system has a unique solution

c. The system has infinitely many solutions

d. The determinant is equal to b

24 [Foundational] • Target: Solution Sets and Free Variables

A consistent system has 5 variables and 3 pivot columns in its coefficient matrix. How many free variables does the system have?

a. 2

b. 3

c. 5

d. 0

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25 [Foundational] • Target: Solution Sets and Free Variables

When does a linear system $A\mathbf{x} = \mathbf{b}$ possess a unique solution?

a. When the system is consistent and every column of A contains a pivot (no free variables)

b. Whenever the number of equations equals the number of variables

c. Whenever $\mathbf{b} \neq \mathbf{0}$

d. When there is at least one free variable

26 [Foundational] • Target: Solution Sets and Free Variables

If a system is consistent and has at least one free variable, how many solutions exist over $\mathbb{R}$?

a. Infinitely many

b. Exactly one

c. None

d. Finite number equal to the number of free variables

27 [Foundational] • Target: Homogeneous Systems and Null Space

What is always true about the homogeneous linear system $A\mathbf{x} = \mathbf{0}$?

a. It is always consistent, because $\mathbf{x} = \mathbf{0}$ is always a solution (the trivial solution)

b. It has no solutions if $\det(A) = 0$

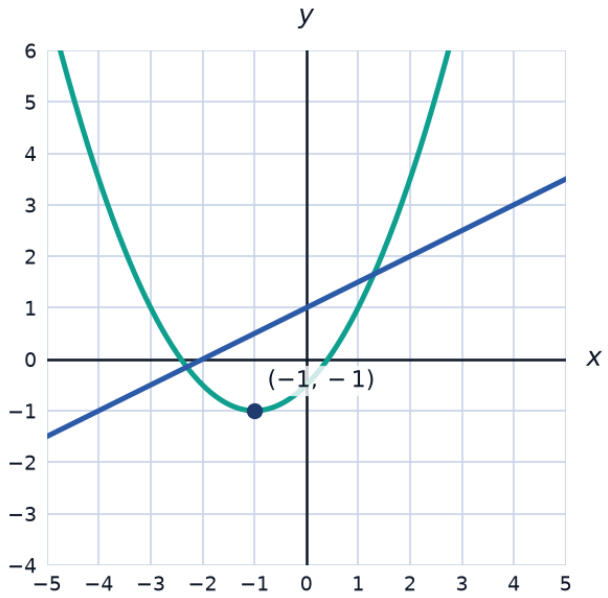
c. It always has infinitely many solutions

d. It only has solutions when A is square

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28 [Foundational] • Target: Homogeneous Systems and Null Space

Under what condition does the homogeneous system $A\mathbf{x} = \mathbf{0}$ have non-trivial solutions (solutions where $\mathbf{x} \neq \mathbf{0}$)?



a. If and only if the system has at least one free variable

b. If and only if A has more rows than columns

c. If and only if all entries of A are positive

d. Whenever $\mathbf{b} = \mathbf{0}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Homogeneous Systems and Null Space

The set of all solutions to $A\mathbf{x} = \mathbf{0}$ forms a subspace of $\mathbb{R}^n$ called the:

a. Null space (or kernel) of A

b. Column space of A

c. Row space of A

d. Span of $\mathbf{b}$

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30 **[Intermediate]** • **Target: Matrix Invertibility and Systems**

According to the Invertible Matrix Theorem, for a square $n \times n$ matrix A, which statement is equivalent to A being invertible?

a. The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$

b. The columns of A are linearly dependent

c. $\det(A) = 0$

d. The null space of A contains non-zero vectors

31 **[Intermediate]** • **Target: Matrix Invertibility and Systems**

If A is invertible and $A\mathbf{x} = \mathbf{b}$, how is the unique solution expressed algebraically?

a. $\mathbf{x} = A^{-1}\mathbf{b}$

b. $\mathbf{x} = \mathbf{b}A^{-1}$

c. $\mathbf{x} = \frac{\mathbf{b}}{A}$

d. $\mathbf{x} = A^{T}\mathbf{b}$

32 **[Intermediate]** • **Target: Matrix Invertibility and Systems**

If A is a 4×4 matrix with rank 3, how many solutions does $A\mathbf{x} = \mathbf{0}$ have?

a. Infinitely many

b. Exactly one

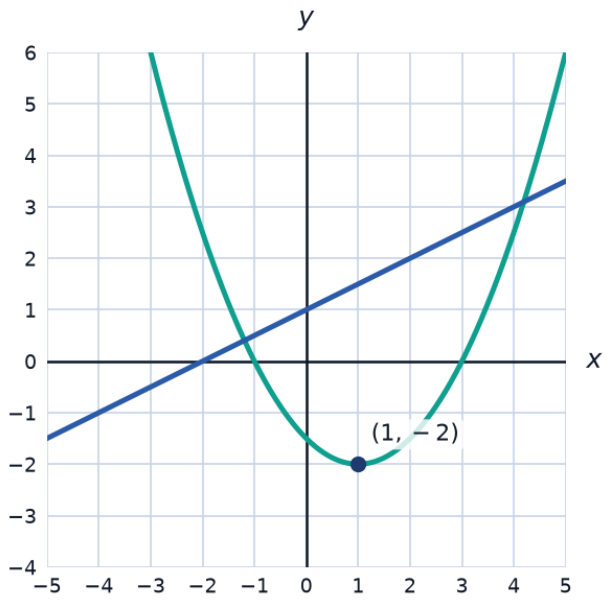
c. None

d. 3

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33 [Intermediate] • Target: Matrix Invertibility and Systems

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?



a. 1

b. 2

c. 3

d. 4

34 [Intermediate] • Target: Setting up the augmented matrix

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x?

a. 1

b. 2

c. 3

d. 4

35 [Intermediate] • Target: Elementary row operations

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation?

a. 1

b. 2

c. 3

d. 4

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36 [Intermediate] • Target: Solving 2x2 systems by elimination

Solve for z: $x + y + z = -8$, $y + z = 8$, $z = -4$ (already row-echelon).

a. -16

b. 12

c. -4

d. -2

37 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Solve for z: $x + y + z = -5$, $y + z = -4$, $z = -2$ (already row-echelon).

a. -1

b. 0

c. -4

d. -2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Row echelon form and back-substitution • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve for x: $x + y + z = -4$, $y + z = 4$, $z = -7$ (already row-echelon).

a. 11

b. -7

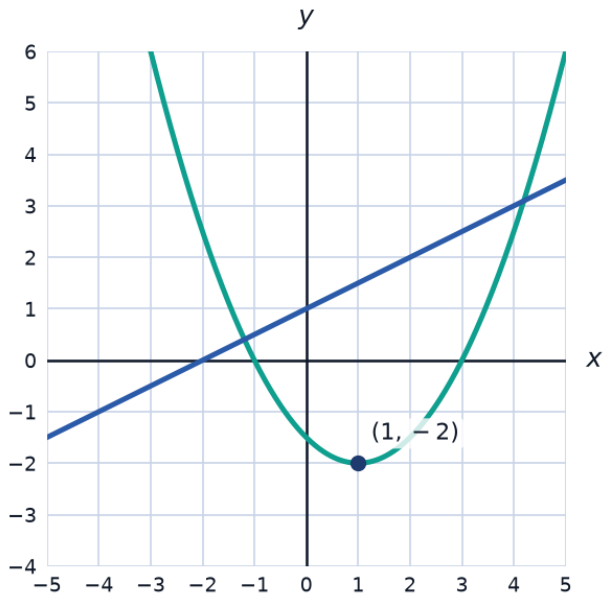
c. -8

d. -6

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39 [Challenging] • Target: Inconsistent and dependent systems • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Solve for y: $x + y + z = 8$, $y + z = -7$, $z = -3$ (already row-echelon).



a. -4

b. 15

c. -3

d. -2

40 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 4.01 Determinants

Solve for y: $x + y + z = 7$, $y + z = -7$, $z = -5$ (already row-echelon).

a. 14

b. -5

c. -2

d. 0

41 [Challenging] • Target: Systems with a parameter • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve for z: $x + y + z = 3$, $y + z = -8$, $z = 4$ (already row-echelon).

a. 11

b. -12

c. 4

d. 6

42 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Solve for z: $x + y + z = 4$, $y + z = -2$, $z = -1$ (already row-echelon).

a. 6

b. 1

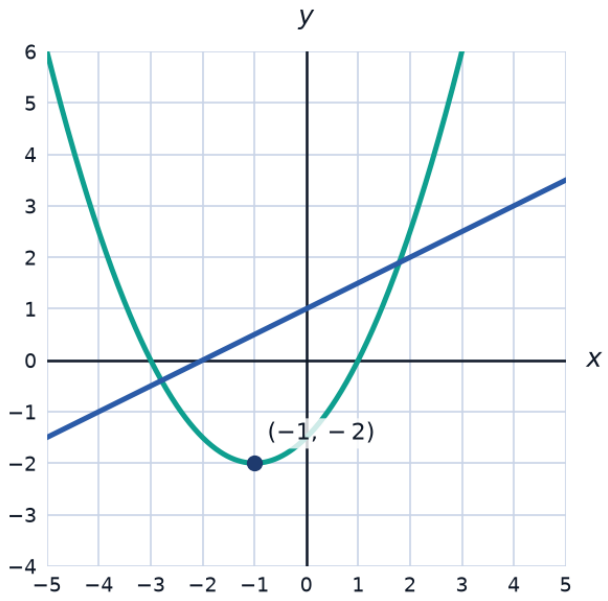
c. -1

d. 0

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43 [Challenging] • Target: Augmented Matrices and Row Echelon Form (REF) • Cross-Lesson: 4.01 Determinants

Solve for y: $x + y + z = 2$, $y + z = -8$, $z = 5$ (already row-echelon).



a. 10

b. -13

c. 5

d. -11

44 [Challenging] • Target: Solution Sets and Free Variables • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve for y: $x + y + z = 6$, $y + z = -4$, $z = 6$ (already row-echelon).

a. 10

b. -10

c. 6

d. -8

45 [Challenging] • Target: Homogeneous Systems and Null Space • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Solve for z: $x + y + z = -4$, $y + z = -1$, $z = -1$ (already row-echelon).

a. -3

b. -1

c. 0

d. 1

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